

FELIX LI

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Engineering, Co-op

Graduation Date: 2027

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Databases, Compilers, Systems Programming and Concurrency

TECHNICAL SKILLS

Frontend: React, Next.js, Tailwind CSS, JavaScript, TypeScript

Backend: C#, ASP.NET Core, .NET MVC, Flask, Python, Java, C++, SQL

Machine Learning & Data: Pandas, Scikit-Learn, Matplotlib, Selenium

AI & LLM Tools: Pinecone, Voyage AI, Anthropic Claude API

Cloud/DevOps: Google Cloud Platform, AWS, PostgreSQL, MySQL, Postman, Git

EXPERIENCE

AI Applications Developer

Jul - Aug 2025

Encore Financial

Toronto, ON

- Built a robust AI-powered financial assistant by integrating **Anthropic's Claude API** with a custom **Node.js/Express** orchestration layer and **React** frontend, enabling real-time queries, document uploads, and dynamic context injection to enhance user interaction and system extensibility.
- Improved chatbot response accuracy by implementing a **Retrieval-Augmented Generation (RAG)** system using **Voyage AI embeddings** and **Pinecone vector database**, enabling high-precision semantic search across complex financial documents.
- Engineered advanced **LLM** workflows with **dynamic prompt construction** and **context management** strategies, enabling accurate **multi-turn interactions** and domain-specific reasoning in a financial chatbot application.

Software Developer

Jan - Apr 2025

titanium.agency

Waterloo, ON

- Reduced page load times by **62%** through SQL query and stored procedure optimization, including targeted indexing, removing redundant computations, and simplifying aggregations.
- Engineered a data archival workflow to transfer historical records from **MySQL** to partitioned **BigQuery** tables, reducing cloud costs by **30%** and improving query performance for transactional workloads.
- Developed real-time automation algorithms using **C#** and **Flask** to optimize budget adjustments and ad pausing through API integrations, leveraging **GCP BigQuery**, **Cloud SQL**, and **MySQL** for historical data analysis, enhancing data-driven decision-making.
- Designed and implemented an automated **ETL** pipeline using **C#** to ingest advertising performance data from third-party marketing APIs, perform data transformation and enrichment, and load structured results into **MySQL**, enabling real-time ad campaign analytics via integrated Tableau dashboards.

Undergraduate Research Assistant

May - Aug 2024

University of Waterloo

Waterloo, ON

- Developed a **Python**-based pipeline to analyze over **2 million** log entries from **12** software systems, generating ground truths for **template-based log parsing** and **saving 10+ hours weekly** by automating large-scale log data analysis.

Full Stack Developer

Sept - Dec 2023

RIFO Holdings Group

Toronto, ON

- Enhanced realtor efficiency by processing and visualizing **real-time** housing data from **2.5M+** Canadian listings using **React**, **ASP.NET Core**, and **MySQL**, enabling seamless front-back integration via **RESTful APIs**.
- Reduced data processing time by **95%** through parallel and asynchronous programming within API workflows, improving backend performance and enabling responsive user experiences.

PROJECTS

NBA Match Predictor - 🔄

Python, Pandas, Scikit-Learn, Matplotlib

- Trained and optimized a Random Forest classifier using **RandomizedSearchCV** and 5-fold cross-validation, achieving **66%** accuracy in predicting winning teams and identifying key predictive features via importance scores.